

CycloProp: a cyclorotor propulsion module

PUSHPAK Grand Challenge, Stage 1 preliminary design report

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1 Submission identity

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2 Summary

One cycloidal rotor. 3 blades on a 110.0 mm radius, NACA 0020, pitching plus or minus 40 degrees about an axis at 30 percent chord, turning at 2405 rpm, driven by a single outrunner through a toothed belt at 3.5 to 1. Design thrust is 18.0 N against a requirement of at least 10 N.

The module weighs 607.97 g at the nominal budget and 684.70 g in the conservative column, so thrust to weight is 3.018 at the design point. Four cases are reported rather than one, because reporting a single case is how this design misread itself for a fortnight early on, and the worst of the four stacks a low thrust coefficient on the conservative mass and still gives 2.5457.

Blade pitch is passive. One four-bar per blade, all three sharing a single offset pivot 15.4 mm from the rotor axis, and thrust vectoring is the direction of that offset rather than a separate mechanism. Two 12.5 g servos turn a phasing carrier through a 1.5 step up and reach 120 degrees of vector authority.

What this report is not: it is not a measurement, and it is not CAE. No CFD has been run, no finite element model exists, nothing has been built and no coupon has been tested. Every load is closed form, every allowable is a published class value and the thrust coefficient is transferred from somebody else's rotor. Stage 1 asks for a preliminary design and this is one. The sections that would be weak if it claimed more are marked, and the claims table near the end carries every headline number with its confidence and its failure mode.

3 Cyclorotor concept and configuration

One rotor, not a cluster. The blades run parallel to the rotation axis and pitch cyclically, so the rotor makes a resultant force that can be pointed anywhere in the plane normal to its axis without moving anything on the airframe.

The single rotor was chosen against redesigned two and three rotor clusters on one common model. Same thrust coefficient, same figure of merit, same efficiency chain, same mass build-up, same module boundary, same drive shortlist. Each layout was swept over radius in small steps and its best conservative thrust to weight kept, and every contested assumption was set in the cluster's favour: wakes that never overlap, no interaction penalty, one shared motor sized on total power, one shared controller, and the same coefficient for all three even though splitting the thrust drops the per rotor Reynolds number.

Layout	Per rotor thrust	Rotor speed	Largest dimension	Module mass	Stacked T/W
single, 3 blades	18.0 N	2405 rpm	290.4 mm	580.05 g	2.517
two rotors	9.0 N	3215 rpm	400.0 mm	786.12 g	1.855
three rotors	6.0 N	3976 rpm	490.0 mm	984.94 g	1.479

The single rotor wins the decision metric by 36 percent and it wins every other column too. The reason is not subtle once the mass lines are sorted: splitting the thrust splits the aerodynamics and does not split the hardware, since seven of the thirteen envelope lines multiply by the rotor count while only three are shared. Total blade area also rises, because smaller rotors run at lower tip speed for the same per rotor thrust. Chord Reynolds falls with the split too, from 134,074 on the single rotor to 94,805 and 77,408, which takes both cluster rows below the measured band and the single rotor stays inside it.

Those three rows are built on the week 2 mass envelope, applied identically to all three layouts. The refined budget later in this report moves the winning row from 2.517 to 2.5457 and leaves the other two where they are.

The module boundary is taken from the problem statement rather than chosen: blades, frame, pitch mechanism, motor, actuator and mounting hardware. Two disputed allocations were settled before anything was scored, and both push our own number down. ESCs count in full, because the motor does not turn without one. Mounting counts in full, so every fastener, insert, lug and bonded joint.

Three more configuration choices, each of which could have gone the other way. Three blades rather than four or six, because solidity is the quantity held fixed and at fixed solidity fewer blades give more thrust, which is Benedict's own finding and matches Kellen's reported optimum for this Reynolds band. NACA 0020 rather than something thinner, because thick sections stay efficient at this scale and a thick section is what makes the blade buildable as a closed cell with a real spar inside it. One motor with a belt reduction rather than direct drive, because rotor torque at the design point is 1.41944 Nm at 2405 rpm and no outrunner in this mass class makes that directly.

4 Preliminary rotor sizing

The shape family is Kellen's UAV scale optimum: 3 blades, chord at 0.66 of the radius, blade aspect ratio 4, NACA 0020, pitching plus or minus 40 degrees. Solving that family for a given thrust pins the Reynolds number whatever the radius, because chord and span both scale with radius and the size cancels out of the product of tip speed and chord.

Parameter	Value
Radius	110.0 mm
Chord	72.6 mm
Span	290.4 mm
Blades	3
Airfoil	NACA 0020
Pitch amplitude	40 degrees about 30 percent chord
Rotor speed	2405 rpm

Parameter	Value
Tip speed	27.70 m/s
Chord Reynolds	134,074
Solidity	0.3151

Solidity sits inside the 0.30 to 0.40 band Kellen measured this shape family across, and the design point sits inside the 100,000 to 300,000 Reynolds range of the same study, with his own optimum rotor at 186,000. That matters more than it sounds. The earlier support for carrying a thrust coefficient across a Reynolds change was Shrestha and Benedict, whose invariance range stops at 100,000, so the design point used to sit past the end of an argument made about something else. It now sits inside a measured band on its own shape.

The blade area coefficient is 0.6055 and it is held deliberately below every measured or corrected value available. Kellen's optimum rotor of this shape family gives 0.6648, measured. Benedict's quad hover point, recomputed from the printed pages rather than from the pairing this project used to carry, gives 0.7211. His twin gives 0.8114 at a solidity far outside the band, so that one is an upside indication and nothing rests on it. Raising the nominal to any of the three would raise thrust everywhere and spend margin the design does not need.

The conservative coefficient is 0.5752, the nominal less 5 percent for blade flexibility and nothing else. It used to carry a further 10 percent for a configuration transfer that changed blade count, airfoil and chord ratio at once with nothing measured behind it. Kellen's measurement on this shape family retired that half. What is left is an engineering downside scenario, not a published lower bound, and it is labelled that way in the evidence ledger.

Radius. Inside a fixed shape family at fixed thrust, aerodynamic power falls roughly as one over radius while rotor torque rises with it, and every geometry scaled mass line grows as the square or the cube. Conservative thrust to weight therefore rises all the way down the radius range and the best row is not the one the design uses. What picks 110.0 mm is the drive. A smaller rotor wants a belt ratio the selected motor cannot spin to on a 6S pack, and the motor that has the speed does not have the torque. So the radius is the smallest one a named drive holds continuously, which is a real finding and points at the cheapest available improvement.

5 Blade arrangement and pitch-control concept

Three blades at 120 degree spacing, each on its own passive four-bar, all three sharing one offset pivot. No blade carries an actuator. The offset link's length sets the pitch amplitude and its direction sets the phase, so one linkage answers the pitching requirement and the vectoring requirement together.

Link	Length	What it is
L1 rotor arm	110.0 mm	input crank, frozen by the radius
L2 offset link	15.4 mm	ground link, solved for the pitch amplitude
L3 pitch link	105.0 mm	output crank
L4 pitch horn	24.4 mm	coupler

Only L2 was solved. Bisecting on peak to peak pitch travel gives 15.4 mm for plus or minus 40 degrees. L1 is the radius, and L3 and L4 come off a vehicle Kellen built, with L3 rescaled by a sweep rather than by his ratio: his length scaled to this radius costs more than twice the carrier torque and leaves the two servos holding on exactly half their stall figure, which is not a margin.

Grashof and the assembly mode. Sorted, the links are 15.4 mm, 24.4 mm, 105.0 mm and 110.0 mm. The shortest plus the longest is less than the other two added together, and the shortest link is the ground, so this is a double crank: the rotor arm turns fully, which it has to, and the pitch link turns fully about the offset pivot once per rotor revolution.

Transmission angle runs 58.58 to 143.23 degrees. The conventional band is 40 to 140 and the obtuse end sits outside it, so the cost is stated rather than the band quoted: the worst sine over the revolution is 0.599, meaning the mechanism's poorest moment arm is 60 percent of its best. The nearest approach to a dead centre is 36.8 degrees away, so nothing is near a singularity.

The solved schedule is 37 rows at 10 degree spacing. It reaches 40 degrees both ways, closes exactly on itself over the revolution, and lags the offset direction by 11.00 degrees. Fitting a cosine to it leaves an rms residual of 1.1951 degrees, which is 3 percent of amplitude. That residual is not noise. It is where the side force comes from, and a prescribed sinusoid does not have it.

One geometric result set the whole drivetrain layout. The pitch link passes within 0.004 mm of the rotor axis, which is a crossing rather than a near miss, and it follows from the link set rather than from bad luck. So the plane the pitch links sweep cannot contain the rotor shaft. The shaft stops inboard of that plane, the rotor is driven from one end only, and the offset pivot is fed by a post from a phasing carrier sitting outboard of everything that rotates with the rotor. Clearance to the next blade is not close at 94.6 mm.

Thrust vectoring. The command is the direction of the offset link, and rotating it rotates the whole pitch schedule rigidly. Two servos of the 12.5 g class drive a phasing carrier through sector gears set 180 degrees apart, which doubles the holding torque and preloads the mesh so backlash does not appear as thrust direction error. The servo gear is 60 mm and the carrier ring gear is 40 mm, a 1.5 step up, so 80 degrees of servo travel becomes 120 degrees of carrier.

Phase command	Vertical	Lateral	Resultant	Direction
-60 degrees	9.0 N	-15.5885 N	18.0 N	-60 degrees
-30 degrees	15.5885 N	-9.0 N	18.0 N	-30 degrees
0 degrees	18.0 N	0 N	18.0 N	0 degrees
30 degrees	15.5885 N	9.0 N	18.0 N	30 degrees
60 degrees	9.0 N	15.5885 N	18.0 N	60 degrees

Direction follows command one to one and the magnitude holds across the range. Be clear about why that comes out so clean: the rotor is axisymmetric, the blades are evenly spaced and the inflow in this model is a uniform vector, so rotating the command rotates the entire solution exactly. **That table is a statement about the model's symmetry and about what the mechanism can reach. It is not a measurement of force at any of those commands.** What breaks it in hardware is everything the model leaves out, and none of it is symmetric.

Holding torque peaks at 0.1389 Nm at three per revolution, 120 Hz, which is above any servo's control bandwidth. Through the step up and across two servos that is 0.0463 Nm each against half

of a 0.216 Nm stall figure, a margin of 2.33, and the phase jitter is bounded by gear backlash at 0.1432 degrees of carrier rather than by the servo loop.

Side force is the open risk in this section and it is stated as one. The model puts the resultant 11.978 degrees round from the offset direction, of which 11.00 comes from the linkage phase delay and only 0.978 from the aerodynamics. Measurement says the aerodynamic part is much larger. Sirohi measured about 10 degrees, Adams 15 to 35 depending on amplitude and rotor speed, and Benedict's twin sat at 30 degrees at its operating point. A quasi steady model with uniform inflow has no wake return, no shed vorticity and no dynamic stall hysteresis, and all three feed the lateral component, so under-prediction is the expected failure and not a surprise.

The tilt is a bias rather than a loss, and the mechanism absorbs a known bias for free by indexing the carrier zero at assembly. What the uncertainty costs is vectoring authority: indexed at the centre of the measured band, the worst case residual comes out of the 120 degrees and leaves the module with roughly four fifths of it. That is still a usable vectoring range and it is the single largest consumer of authority in the design.

6 Estimated thrust and power requirement

Design thrust is 18.0 N and the conservative case is 17.0992 N. Both clear the 10 N requirement on their own. Thrust comes from the blade area coefficient route, $T = Ct \times 0.5 \times \rho \times u^2 \times N \times c \times s$, at a tip speed of 27.70 m/s over a blade area of 0.06325 square metres.

Choosing 18.0 N is arithmetic rather than ambition. The conservative case has to clear 10 N by itself, which puts a floor under the nominal. Above that, the geometry scaled mass lines do not care what thrust the rotor is turning for, so extra thrust buys mass ceiling while only the drive grows. The trade runs out at 18.0 N, and it runs out because of the drive rather than the physics.

Design thrust	Mass ceiling at T/W 2.5	Ideal power	Rotor speed	Rotor torque	Drive consequence
13.0 N	530 g	118.474 W	2044 rpm	1.0251 Nm	the roomiest row, 12.65 A of 20.8 A
16.0 N	652 g	161.766 W	2267 rpm	1.2617 Nm	17.27 A of 20.8 A
18.0 N	733 g	193.026 W	2405 rpm	1.4194 Nm	the design point, 20.61 A of 20.8 A
20.0 N	815 g	226.075 W	2535 rpm	1.5772 Nm	nothing in the shortlist fits

At 20.0 N the motor input is still inside the continuous power rating, so it is not a power limit. It is torque and speed together: the rotor wants a ratio above 4 and a KV450 on 6S cannot spin to the motor speed that implies.

Power is closed three ways rather than asserted once.

Momentum theory over the projected frontal area, 2R times span, gives an ideal induced power of 193.026 W for 18.0 N. No rotor beats that. At Kellen's measured figure of merit of 0.6 the blade

aerodynamic power is 321.71 W. An independent route through Benedict's measured power loading puts the same thrust at 290.323 W, so two routes that share no equation agree to 9.8 percent. Induced velocity is 10.7237 m/s against a tip speed of 27.70, an inflow ratio of 0.3871, and that is high enough to be the main reason a simple model should not be trusted for magnitude.

Term	Value	Where it comes from
blade aerodynamic power	321.71 W	ideal power over a figure of merit of 0.6
rotor tare	35.746 W	10 percent of shaft power, measured on a flight weight rotor
motor input power	457.573 W	shaft power through a 0.93 belt and a 0.84 motor
electrical power at the ESC input	481.655 W	motor input over 0.95
actuator draw	6.0 W	two servos holding against residual link load
controller draw	2.0 W	offset controller board
module electrical power	489.655 W	the three above

The three efficiencies are assumed rather than measured and they are the only unevidenced links in the chain. The motor figure is the sensitive one.

The drive, on a derated continuous rating. The selected motor is a T-Motor Antigravity MN5006 KV450 at 106 g. Its published figures are a maximum power and a peak current over 180 seconds, which is a three minute maximum and not a hover rating, so every rating in the shortlist is derated by 0.80 before anything is selected against it. Nothing justifies 0.80 rather than 0.70 or 0.90 except ordinary practice, and the problem statement states no endurance requirement to size it against. It is an assumption and it is marked as one.

What can be done without a source is bound the consequence. The design draws 0.704 of the published 180 second power and 0.7927 of the published 180 second current, and that second figure is also the derate at which the selection breaks even: above it the motor covers the design point and below it it does not. Neither exit from that is open. Backing the design point down a row fails because the stacked thrust to weight case needs 17.6768 N and no lower row of the sensitivity table reaches it, and a larger motor fails because motor mass is a power class item against 12.5 g of conservative headroom. What makes it tolerable is the duty rather than the margin: 0.7927 is a fraction of a three minute rating, so for a demonstration inside three minutes there is a fifth of the datasheet figure in hand. A dynamometer run early in Stage 2 settles it, and it is the first drive gate in the plan.

A drive is accepted only if three things hold at once. Power: 457.573 W of a derated 520.0 W continuous. Torque: 0.4361 Nm against 0.4414 Nm continuous at 3.5 to 1 through a 0.93 belt, which is the tight one at 99 percent. Speed: the motor turns 8417 rpm and a 6S pack can reach 9433.5 against its internal resistance at the working current. Four other motors were screened and each one fails at least one of the three.

Azimuthal load distribution. A cycle averaged coefficient hides what a blade actually sees, so the load model runs 36 azimuths with a uniform induced inflow and the solved pitch schedule rather

than a prescribed sinusoid. Peak vertical force per blade is 15.0061 N against a cycle mean of 6.0 N, so peak to mean is 2.501. That sits below the published 3 to 4 range, and the honest reading is that the model under-predicts the peak rather than that this rotor is gentler than the literature. Structure is sized on 4.0 regardless.

The cycle mean lateral force is trimmed to zero by pointing the offset off module vertical. The instantaneous lateral force still reaches 10.0987 N per blade inside the revolution, which is a bearing and frame load rather than a thrust loss.

7 Estimated module weight and thrust-to-weight ratio

The module is 33 budget lines, every one a drawn section, a catalogue part or a stated allowance, and every one pointing at the coarse envelope line it refines. The budget is built by a script from the section drawing and the material densities, so the arithmetic reruns rather than being typed.

Line	Nominal	Conservative	Class
blade foam cores, 3 off	28.79 g	33.11 g	calculated
blade skins, 3 off	29.08 g	33.44 g	calculated
blade spar tubes, 3 off	17.42 g	20.03 g	calculated
blade root close-outs, 3 blades	19.95 g	22.35 g	machined
rotor spider arms, 6 off	21.48 g	24.71 g	calculated
rotor hub bosses, 2 off	14.41 g	16.14 g	machined
root attachment brackets, 6 off	18.0 g	20.16 g	machined
pitch bearings, 6 off	7.8 g	8.42 g	catalogue
pitch links with rod ends, 3 off	12.28 g	13.27 g	catalogue
pitch horns, 3 off	6.58 g	7.37 g	machined
offset pivot post and pin	6.5 g	7.28 g	machined
phasing carrier ring, 40 mm gear	7.9 g	8.85 g	machined
servo sector gear, 60 mm	4.5 g	5.04 g	machined
carrier support bearings, 2 off	4.4 g	4.75 g	catalogue
rotor shaft tube	36.01 g	41.41 g	calculated
shaft end plugs, 2 off	17.24 g	19.31 g	machined
main bearings, 2 off	16.0 g	17.28 g	catalogue
bearing blocks, 2 off	16.0 g	17.92 g	machined
frame tubes, 4 off	46.36 g	53.31 g	calculated
motor mount plate	8.0 g	8.96 g	machined
airframe mount lugs, 4 off	7.2 g	8.06 g	machined
frame and mount design reserve	15.0 g	18.75 g	allowance
motor, MN5006 KV450	106.0 g	114.48 g	catalogue
rotor belt pulley, 56 tooth	28.15 g	31.53 g	machined
motor belt pulley, 16 tooth	6.62 g	7.41 g	machined
drive belt	10.2 g	11.02 g	catalogue
belt tensioner and bracket	6.0 g	6.72 g	machined
esc, 40 A 6S class	19.5 g	21.06 g	catalogue
vectoring actuator servos, 2 off	25.0 g	27.0 g	catalogue
pitch offset controller	8.5 g	9.18 g	catalogue
module wiring harness	15.6 g	19.5 g	allowance

Line	Nominal	Conservative	Class
fasteners and threaded inserts	14.0 g	17.5 g	allowance
structural adhesive at module joints	7.5 g	9.38 g	allowance
module total	607.97 g	684.7 g	

The growth rate is a property of the line, not of the module. Each line is sorted into one of four classes and each class carries a rate, decided before the total was looked at: 8 percent for a catalogue part, 12 for a drawn and machined one, 15 for a section computed from stock, 25 for anything not drawn at all. Weighted across the module that is 12.6 percent. Every group also has to land within 25 percent of the coarse envelope line it refines, so the budget cannot quietly move mass from one part of the module into another, and the four groups that moved more than 10 percent each have a stated reason.

There is a visible 15.0 g reserve, and it sits on the frame and mount group because that is the least developed part of the module. It covers gussets, cable clamps, the servo bracket and the ESC tray, none of which is drawn. It is not spread across rounded lines and it is not buried inside a growth rate.

What is genuinely fixed in this module is one 8.5 g controller board. That is worth saying plainly, because the usual thrust to weight argument for a bigger rotor is that fixed hardware amortises over more thrust. Sorted honestly, the wiring follows the envelope, the fasteners follow the frame, the servo torque follows the pitch link load which follows thrust, and the bearings, shaft and transmission follow rotor torque. Almost nothing here is free when the rotor grows.

Case	Thrust	Mass	Weight	T/W
design point	18.0 N	607.97 g	5.9642 N	3.018
mass downside alone	18.0 N	684.7 g		2.680
coefficient downside alone	17.0992 N	607.97 g		2.867
both stacked	17.0992 N	684.7 g		2.5457

All four clear the requirement. The stacked row is the one that matters, since it applies the low coefficient and the conservative mass at the same time, and it clears by 12.5 g of mass. That is a real pass and a thin one, and it is described as thin rather than rounded up.

An internal target of 2.75 on the stacked case was set early as a margin goal. It is not met, it is not claimed anywhere, and no further pass over the budget closes it: every remaining line is a drawn section or a catalogue part, and trimming one to reach a number is exactly the move the budget rules exist to stop. The two routes that would close it are a lower KV motor on more cells and a measured thrust coefficient, and both are Stage 2 work.

One trade was priced and declined. Balancing the blade chordwise would lift the pitch link margin from 3.30 to 4.69, and the nose ballast that needs across three blades takes the stacked case to 2.406, under the hard limit. Spending a requirement to improve a margin that already passes twice over is the wrong trade, so the blade stays unbalanced and the load path carries it.

8 Initial material and manufacturing approach

Six materials carry load and each one was chosen because a structural calculation needed an allowable. They are defined once, in the solver, and both the structures work and this section cite that definition rather than restating it.

Material	Where	The margin it decides
Rohacell 51 IG class PMI foam, 88 percent fill	blade core	blade bending, through skin wrinkling
60 gsm carbon twill, 2 plies roll wrapped CFRP tube	blade skin spar, pitch links, rotor shaft, frame tubes	blade bending, 2.83 and 1.97 shaft torsion 17.58, combined 9.44
7075-T6 aluminium	horns, root fittings, brackets, blocks, lugs	pitch link path, 3.30
6061-T6 aluminium	pulleys, carrier ring gear, sector gear	none, these are stiffness parts
Araldite 2011 class epoxy paste	shaft plugs, root fittings, block bonds	none yet, and that is stated

The foam is structural here and not a filler. Skin wrinkling over a soft core is what limits the blade, and the wrinkling stress depends on the skin modulus and both core moduli, two of the three belonging to the foam. Drop to a lighter grade and the wrinkling stress falls by roughly a quarter, which takes the blade's combined margin at overspeed under its floor. The grade is part of the structure and a substitution means a recalculation.

8.1 Structural design and margins

This is the 15 percent structural criterion and it belongs with the material choice, because each allowable is what makes a margin real.

The headline is that this rotor is a centrifugal machine before it is an aerodynamic one. Each blade pulls 221.464 N radially at the design point, and at a declared 1.20 overspeed that becomes 318.908 N, because centrifugal load goes as the square of speed. Mean aerodynamic force per blade is 6.0 N and structure is sized on 4.0 times that. The ratio between the two loads is 9.2 here, where Runco measured 4.4 on a much smaller rotor.

Case	Demand	Allowable	Margin
blade bending, aerodynamic only	0.8712 Nm	25.2528 Nm	28.99
blade bending, aerodynamic and centrifugal		25.2528 Nm	2.83
blade bending at 1.20 overspeed		25.2528 Nm	1.97
rotor shaft torsion	1.41944 Nm	24.9563 Nm	17.58
rotor shaft, bending and torsion combined	2.22965 Nm of bending		9.44

Case	Demand	Allowable	Margin
pitch link path, the horn governs	105.93 N	349.727 N	3.30
blade attachment, centrifugal	221.464 N	540.0 N	2.44
blade attachment at 1.20 overspeed	318.908 N	540.0 N	1.69

The spread is the point. Two margins sit under 2 and everything else is over 3, so the module is sized by the blade in combined bending and by the blade attachment, and Stage 2 effort belongs at those two joints rather than at the shaft or the frame. The blade's own section allowable is set by skin wrinkling over the foam and not by the laminate, and the spar would take more than twice the section's rating on its own.

The attachment is the tightest joint and the number is not the worst part of it. The two pitch bearings that carry each blade give 540.0 N together and see 318.908 N at overspeed. They oscillate through 80 degrees under a steady load rather than rotating, which is a fretting duty that a static rating describes not at all. It needs a supplier's oscillating derate or a bench test, and it has neither.

Blade stiffness came out fine in bending and interesting in torsion. Tip deflection under the peak blade load is 0.0936 mm and twist under the aerodynamic pitching moment is 0.0145 degrees, so neither is a design driver. The blade is driven in pitch from one end though, so the centrifugal pitching moment on an unbalanced blade has to go through the blade's own torsional stiffness, and 2.2058 Nm winds the far end up by 2.3509 degrees. That is 4 percent of the pitch amplitude, and it is most of the 5 percent blade flexibility allowance the conservative coefficient carries. The allowance had a bound over it before and has a calculation under it now.

The reaction torque path is blade, spider arm, hub, through shaft, main bearing, bearing block, frame tube, mount lug. The rotor shaft carries 1.41944 Nm and the motor shaft 0.43608 Nm, and the airframe sees the rotor figure as a steady moment about the rotor axis whenever the module makes thrust. Both bearing blocks take it, which is why they sit on the frame tubes rather than on side plates, and there are 4 mount points rather than 3 because a three point mount puts that torque into a triangle whose worst leg carries most of it.

8.2 Manufacturing and cost

Every part has a stock form, a process, one tolerance that matters and a joining method. Nothing needs a process a university lab or a job shop in an Indian metro cannot run, and the one part with no Indian stockist is called out.

The blade is CNC profiled foam in two halves with the spar channel cut, two plies wet laid in a machined two part mould and vacuum bagged, cured at room temperature, then two turned 7075-T6 root fittings bonded over the spar and pinned. The shaft plugs are bonded into the tube first and both journals ground afterwards in one setup, because grinding after bonding is what makes them concentric. Three CNC job lines cover six part families between them, so a shop quotes them as batches. The gear pair is the only job needing a cutter nobody local keeps on a shelf.

What gets measured before it spins, because a rotor at 2405 rpm with 221.464 N pulling on

every blade is not a thing to power up hopefully. Blade masses matched across the set, shaft journal run out on the jig, and pitch angle checked against the solved schedule at 12 azimuths by hand, where peak to peak travel is what matters because the rod ends can correct it. Then the rotor turned by hand through two revolutions at both ends of servo travel, watching for a link going over centre, since the worst transmission angle is 143.23 degrees and it wants feeling rather than assuming. Then a static pull on one attachment above the overspeed load of 318.908 N, and a first spin staged in four steps with current logged against prediction.

Bought parts and material come to 36970 INR, tooling and fabrication to 28800, and the module totals 65770 INR at a longest single lead of 4 weeks. **These are indicative prices at distributor list level and they are not obtained quotations.** No supplier was contacted, and the source column of the full bill of materials names the distributor a part would be bought from rather than one that has quoted for it. Five lines above 4500 INR carry 52 percent of the total and those are what Stage 2 has to replace with written quotes.

The schedule driver is a 4 week foam import with no Indian source. Everything on a 3 week lead sits inside that window, so ordering the foam first is the whole mitigation. If it slips, a lighter core is not a drop in substitute, because the wrinkling calculation depends on the core moduli, and the fallback is a thicker skin and a rerun.

9 Team capability and execution plan

This section is deliberately incomplete and the reason is worth stating. The people, the institution, the prior projects and the tool licences behind this entry are facts about a team, and no part of the automated work that produced this report may write them. The structure below is real, the gap analysis is real, and the fields marked [P-n] are for a person.

Roster: [P-1] [P-2]. One person as of 27 August 2026. Team size is capped at 5 and nothing in Stage 1 needed more than one, though the two solver workstreams in Stage 2 are where a second person would first pay for themselves.

The problem statement names seven areas where preference may be given, which is close to a specification for this section, so it is answered area by area including the areas that are not covered.

Preference area	What the Stage 1 work shows	The gap
Rotor design and unsteady aerodynamics	a coefficient traced to primary sources and recomputed, three scenarios with an evidence class each, a momentum floor, two independent power routes and a 36 point azimuthal load model	no unsteady solver has been run
CAD and mechanical design	dimensioned layout, a swept envelope from the linkage solution, an interface table and a mount pattern	no solid model exists

Preference area	What the Stage 1 work shows	The gap
Kinematic analysis of mechanisms	a four-bar closed per blade, an offset solved by bisection, a schedule that closes on itself, Grashof and transmission angle, and a force vector map	closed form and planar, no multibody model
CFD, FEA and multibody dynamics	none of the three	the largest single gap in the project
Lightweight structures and material selection	six materials each tied to the margin it decides, a section integrated from its own ordinates, 8 margins and a declared overspeed	published class allowables, no coupon test
Motor, actuator and control selection	a five row drive shortlist screened on power, torque and reachable speed, all on derated continuous ratings	four rows and the servo are supplier listings, and no control loop is designed
UAV subsystem integration and testing	interface definitions, an assembly order and six pre-spin measurements	nothing has been built or tested

Execution plan for Stage 2. The window is 3 October to 2 December 2026 and this plan submits on 1 December. All 11 Stage 2 items are scheduled below with the gate that closes each one, and 07-team-and-execution.md carries the same schedule with the tool category, the dependency and the owner against every item.

Weeks	Stage 2 items	Closed when
1 to 2	CAD model of the module, and supplier quotations started	every budget line exists as a solid and the model's mass properties reproduce the budget
2 to 3	kinematic model, then motor, actuator, bearing and controller selection	the multibody schedule matches the closed form one and reproduces the pitch link load
4 to 5	aerodynamic analysis for thrust prediction	a converged run at the design point with mesh and timestep independence shown
6	structural analysis of blades, supports, frame, shaft and linkages	the 8 margins reproduce or move, with every difference explained
7	material selection, mass estimate and thrust to weight	cured laminate modulus measured on a panel built the way the blade is built

Weeks	Stage 2 items	Closed when
8	manufacturability and assembly plan, bill of materials, risk assessment	written quotes for the five lines above 4500 INR, and every open risk owned
8 to 9	build and test plan, then the report	a staged spin plan, an instrumented thrust measurement and a place to run it

Quotations sit in week 1 rather than week 8 because the 4 week foam lead gates the build plan. The aerodynamic item is the one most likely to overrun, and its fallback is a 2D transient study at the design azimuth set, reported as what it is rather than as a full solution.

Three of the four capability gaps have a route that needs no new person. CFD has an open source option and a validation case in Kellen's measured rotor, whose shape family matches this design to within 1.1 percent. FEA is wanted for three small parts already sized by beam formulae, so its job is to find where the idealisation was wrong. Multibody has a published answer to reproduce on day one, so it is cheapest and goes first. Testing is the gap with no software route: [P-9] a layup bench, a test frame, a load cell and somewhere safe to spin a rotor.

10 Evaluation criteria map

Where each published criterion is answered in this report.

Criterion	Weight	Where it is answered
Feasibility of achieving 10 N thrust	15%	Estimated thrust and power requirement: thrust from geometry on a transferred coefficient held below a measurement of this shape family
Feasibility of thrust-to-weight above 2.5	15%	Estimated module weight and thrust-to-weight ratio: a 33 line budget and four cases, all clearing the limit
Kinematic design of blade-pitch and thrust-vectoring mechanism	15%	Blade arrangement and pitch-control concept: loop closure, solved schedule, Grashof and the force vector map
Aerodynamic analysis or simulation quality	15%	Estimated thrust and power requirement: momentum floor, figure of merit, a second power route and an azimuthal model

Criterion	Weight	Where it is answered
Structural design and strength assessment	15%	Initial material and manufacturing approach: two load cases, 8 margins and the analyses not done
Manufacturability, material selection and cost realism	10%	Initial material and manufacturing approach: per part process and tolerance, assembly order and a costed bill of materials
CAD quality, integration readiness and packaging	5%	Cyclorotor concept and configuration, with the swept envelope, interfaces and mounts, and no CAD at this stage
Presentation, viva and technical clarity	10%	This report, the claims table below and the examiner questions in appendix A

11 Claims, evidence and risk

Every headline claim, what produced it, how much weight it carries, what would break it and what Stage 2 does about that.

Claim	Where it comes from	Confidence	Main failure mode	Stage 2 validation
Blade area coefficient 0.6055 gives 18.0 N	transferred from a recomputed hover point, bracketed above by 0.6648 measured on this shape family	medium	the transfer is wrong in the unsafe direction, or the model overstates thrust at an inflow ratio of 0.3871	transient CFD, then a load cell run
Module mass 607.97 g nominal, 684.7 g conservative	33 drawn or catalogue lines, growth by line class	medium to high	a wet layup blade comes out heavy, or undrawn frame parts eat the 15.0 g reserve	CAD mass properties, a mould trial, weighed parts
Blade deflection 0.0936 mm, wind up 2.3509 degrees	closed form beam and torsion on the integrated section	medium	cured laminate modulus below the published class value	coupon panel, then FEA

Claim	Where it comes from	Confidence	Main failure mode	Stage 2 validation
Direction follows the vector command one to one at 18.0 N	model symmetry, not measurement	low on magnitude	wake skew, the frame in the flow, the blade meeting its own wake	two axis load cell across the range
Stacked conservative thrust to weight 2.5457	recomputed from geometry and the mass lines	medium	either input moving, since it clears by 12.5 g	Stage 2 items 3, 6 and 7 together
Blade attachment margin 1.6933 at overspeed	static rating over recomputed centrifugal load, with the oscillating duty computed	medium	the bearing sits at 0.5533 of full recirculation, so it wears where it sits and no catalogue figure covers that	run to failure at speed under load, with the flight grease
Peak to mean blade load 4.0	published simulated range, top of it taken	low to medium	the true peak is higher under dynamic stall	measured blade forces, or CFD
Module electrical power 489.655 W	momentum floor, figure of merit, efficiency chain	medium	all three efficiencies are assumed and the motor one is sensitive	bench measurement on the built module
The drive holds the design point continuously	derated ratings against power, torque and speed	medium	the 0.80 derate has no source, it breaks even at 0.7927, and torque sits at 99 percent	dynamometer run at the working current, early enough that a drive change is still affordable
Side force tilt 11.978 degrees	quasi steady model with uniform inflow	low	measurement puts it far higher and it moves with rotor speed	load cell calibration across the speed range
Module cost 65770 INR, longest lead 4 weeks	distributor list prices, nothing quoted	low on price	gear cutting is priced by setup at a quantity of one	written quotations
Material allowables	published typical values per class	medium	no certificate, no coupon, and the skin modulus is what the blade turns on	coupon panel and a bond shear coupon

12 Sources and how they were read

Five evidence classes are used: measured, somebody measured this quantity on this configuration and the figure was read off the source; derived, computed here from figures measured on a different configuration; transferred, a derived value carried across a change of geometry or Reynolds number; summary, only a summary of the source has been read; assumed, no source at all.

Nobody has put this rotor on a load cell. Where this report says measured, it means measured by somebody else on a rotor of this shape family, and the distinction is kept everywhere it matters.

- **Kellen 2019**, MS thesis, Texas A&M. Read as a machine text extract of the full PDF, which is held in the repository along with the hash and a working re-fetch URL. Source of the figure of merit, the solidity band, the Reynolds range and the 0.6648 coefficient measured on this shape family
- **Benedict 2010**, PhD dissertation, University of Maryland. Read the same way. Source of the nominal coefficient's provenance, the corrected quad hover point, the power loading cross check and the rotor tare fraction
- **Sirohi, Parsons and Chopra 2007**, read. **Xisto et al. 2016**, read. Both support the configuration choices rather than any stored number
- **Shrestha and Benedict**, JAHS 2022, summary class. It carries the Reynolds invariance claim and the blade mass scaling argument, and it is cited as a summary
- **Adams et al. 2013** and **Alsabri et al. 2025**, summary class. The side force band and the peak to mean load range come from them and are labelled
- **Runco and Benedict 2023**, open access, re-cut onto this competition's module boundary to give the benchmark this design is argued against
- **Ramsey 2022** and **Heimerl et al.** are **not read**. Neither is cited as evidence anywhere in this report. Ramsey would give a second structural mass anchor, and Heimerl measured instantaneous blade forces across a Reynolds band that brackets this design, which would replace both the peak to mean estimate and the side force angle with measurements

No figure or table has been reproduced from any source. Every published number used here was recomputed into this project's own conventions from figures printed in the source, and the conversion is recorded in the evidence ledger alongside the printed page it came from.

13 Appendix A: examiner questions

Twelve questions this design should expect, answered short. Where an answer exposed a gap, the report was changed rather than the answer smoothed.

1. Your whole design hangs off a thrust coefficient measured on a different rotor. Why should anyone believe 18.0 N? They should believe the conservative case first. The nominal 0.6055 is transferred, and its original provenance was wrong, so it was recomputed from the printed pages and the corrected value is 0.7211, higher than what is used. Kellen then measured 0.6648 on a rotor whose solidity and chord to radius are within 1.1 percent of this one. Three independent values sit above the number the design is built on. That is the argument, and it does not make 18.0 N a measurement. The claim is that the design closes on a coefficient deliberately below the evidence, and the requirement is 10 N.

2. The published benchmark for a module like this is between 1.78 and 2.13 depending on allocation. You claim 2.5457. Why is that credible? Because the benchmark is a 70

gram micro-scale vehicle re-cut onto this boundary, and blade mass per newton is roughly scale invariant while the drive and structure are not. This module makes 18.0 N from one motor, one belt, one shaft and one pitch mechanism, where the benchmark makes a fraction of that per rotor from four sets of hardware. The gap is not a claim to have beaten anybody at the same scale. It is amortisation, and the report is careful about how little of it is real: the genuinely fixed mass in this module is one 8.5 g board.

3. What in the module is actually fixed as the rotor grows? Almost nothing. The 8.5 g controller. Wiring follows the envelope, fasteners follow the frame, servo torque follows the pitch link load which follows thrust, and the shaft, bearings and transmission follow rotor torque. This is stated in the module weight section because it is the weakest part of the usual argument for a large single rotor, and hiding it would be worse than losing the point.

4. Why 110.0 mm and not the radius your own sweep prefers? The sweep prefers smaller. A smaller rotor is lighter on every geometry scaled line and conservative thrust to weight rises all the way down. What stops it is the drive: below this radius the rotor torque wants a belt ratio the KV450 cannot spin to on 6S, and the motor with the speed lacks the torque. So 110.0 mm is the smallest radius a named drive holds continuously. If a lower KV motor on more cells were available, that is where the design would go next.

5. Your motor sits at 99 percent of its continuous torque and the derate has no source. Is that a design or a hope? It is a judgement, stated as one. T-Motor publishes a maximum over 180 seconds, which is not a hover rating, so every shortlist figure is cut to 0.80 before selection. Nothing supports 0.80 over 0.70 or 0.90 except practice, and the problem statement gives no endurance requirement to size it against. At the derated figure the design point takes 457.573 W of 520.0 W and 0.4361 Nm of 0.4414 Nm, so power has room and torque does not. A thermal run on the bench is what settles it, and it is in the Stage 2 plan.

6. Where does the reaction torque go? Blade, spider arm, hub, through shaft, main bearing, bearing block, frame tube, mount lug. The rotor shaft carries 1.41944 Nm and the motor shaft 0.43608 Nm upstream of the belt, and the report says which is which because that is the first thing a reviewer checks. The airframe sees the rotor figure as a steady moment whenever the module makes thrust, both bearing blocks react it, and the mount has 4 points rather than 3 for exactly that reason.

7. Your transmission angle reaches 143.23 degrees, outside the conventional band. How close is this mechanism to a singularity? Not close. The conventional band is 40 to 140 and the obtuse end sits just outside it, which costs moment arm rather than control: the worst sine over the revolution is 0.599, so the poorest moment arm is 60 percent of the best. The nearest approach to a dead centre is 36.8 degrees. The linkage is a double crank by the Grashof test, both cranks turn fully, and the assembly is checked by hand through two revolutions at both ends of servo travel before first spin.

8. Your vector map shows constant 18.0 N at every command. Real rotors do not do that. Is the table a measurement? No, and the report says so in bold in item 3. The rotor is axisymmetric, the blades are evenly spaced and the inflow in this model is uniform, so rotating the command rotates the whole solution exactly. The table is a statement about model symmetry and about the commands the mechanism can reach, nothing more. Wake skew, the frame and shaft supports sitting in the flow on one side, and the blade passing through its own returning wake all break that symmetry and none of them is symmetric. A two axis load cell across the command range is the measurement that replaces it.

9. Is the blade stiff enough, and how do you know without FEA? In bending, comfortably, and the numbers are small enough that FEA would not change the answer: 0.0936 mm of tip deflection under peak blade load and 0.0145 degrees of aerodynamic twist against a 40 degree amplitude. The one that matters is torsion. The blade is pitched from one end, so the centrifugal pitching moment of 2.2058 Nm on an unbalanced blade winds the far end up by 2.3509 degrees, about 4 percent of amplitude. That is most of the 5 percent flexibility allowance the conservative coefficient carries, so the allowance now has a calculation under it instead of being a chosen floor. Driving the blade from both ends would roughly quarter it and costs hardware the mass budget has no room for.

10. Your side force model gives about a degree of aerodynamic tilt and the literature measures tens of degrees. Why publish it? Because it is the honest output of a quasi steady model with uniform inflow, and because the difference is explainable rather than mysterious: that model has no wake return, no shed vorticity and no dynamic stall hysteresis, and all three drive the lateral component. The stored figure is carried as a lower bound and labelled one. The design handles the tilt as a bias rather than a loss, since indexing the phasing carrier zero at assembly costs no actuator range, and the uncertainty in the bias is what costs range. It is the largest single consumer of vectoring authority in the design.

11. What is the tightest thing in the module? The blade attachment at overspeed, at 1.6933, and the duty is worse than the number. Two pitch bearings per blade give 540.0 N against 318.908 N, but they swing through 80 degrees under a steady load rather than rotating, and a static rating says nothing about fretting. That is why the overspeed case is declared at all: at the design point the same joint reads 2.44 and looks comfortable.

12. What is the first test you run in Stage 2? A coupon panel, laid up the way the blade is laid up, for cured laminate modulus and areal mass. It is cheap, it is fast, and the blade allowable turns on the skin modulus more than on anything else, so a 15 percent shortfall there moves the margin that sizes the module. Second is a static pull on one blade attachment above the overspeed load of 318.908 N. The rotor does not spin until both have passed.

14 Appendix B: numbers and provenance

Every number in this report is defined once, in `stage-1/design/numbers.json`, and the prose cites it rather than restating it. A gate script recomputes the physics from the stored geometry and the mass lines, applies the competition limits to what it recomputed rather than to any stored headline, and fails on disagreement. It also audits this document: every number in the narrative carrying a physical unit has to match a computed value in the same dimension.

14.1 Numbers used

- `performance.thrust_N` = 18.0
- `geometry.radius_m` = 0.11
- `results.thrust_to_weight_conservative` = 2.5457
- `geometry.chord_m` = 0.0726
- `geometry.span_m` = 0.2904
- `geometry.blades` = 3
- `geometry.pitch_amplitude_deg` = 40.0
- `geometry.pitch_axis_pct_chord` = 30.0

- operating.rpm = 2404.79
- operating.tip_speed_ms = 27.7012
- operating.reynolds = 134074.0
- performance.thrust_N_conservative = 17.0992
- performance.blade_area_coeff = 0.6055
- performance.blade_area_coeff_low = 0.5752
- performance.blade_deflection_thrust_loss = 0.05
- performance.solidity = 0.3151
- performance.blade_area_m2 = 0.06325
- performance.aero_power_W = 321.71
- performance.ideal_power_W = 193.026
- performance.figure_of_merit = 0.6
- performance.aero_power_W_published = 290.323
- performance.power_spread = 0.0976
- performance.tare_power_W = 35.746
- performance.electrical_power_W = 481.655
- performance.module_electrical_power_W = 489.655
- performance.actuator_power_W = 6.0
- performance.controller_power_W = 2.0
- performance.induced_velocity_ms = 10.7237
- performance.inflow_ratio = 0.3871
- performance.blade_load_peak_to_mean = 2.501
- performance.motor_input_W = 457.573
- performance.motor_rpm = 8416.8
- performance.motor_torque_Nm = 0.4361
- performance.motor_input_current_A = 20.611
- performance.belt_ratio = 3.5
- performance.blade_tip_deflection_mm = 0.0936
- performance.blade_twist_deg = 0.0145
- efficiency.transmission = 0.93
- efficiency.motor = 0.84
- efficiency.esc = 0.95
- pitch.offset_m = 0.0154
- pitch.pitch_link_m = 0.105
- pitch.horn_m = 0.0244
- pitch.phase_delay_deg = 11.0
- pitch.schedule_rms_residual_deg = 1.1951
- pitch.transmission_angle_min_deg = 58.58
- pitch.transmission_angle_max_deg = 143.23
- pitch.axis_keepout_mm = 0.004
- pitch.neighbour_clearance_mm = 94.6
- pitch.pitch_bearing_travel_deg = 80.0
- pitch.vector_range_deg = 120.0
- pitch.phase_authority_deg = 120.0
- pitch.servo_travel_deg = 80.0
- pitch.gear_step_up = 1.5
- pitch.carrier_gear_mm = 40.0
- pitch.servo_gear_mm = 60.0

- pitch.servo_mass_g = 12.5
- pitch.actuator_count = 2
- pitch.carrier_torque_Nm = 0.1389
- pitch.servo_torque_Nm = 0.0463
- pitch.servo_stall_torque_Nm = 0.216
- pitch.servo_torque_margin = 2.332
- structure.carrier_phase_jitter_deg = 0.1432
- pitch.side_force_tilt_deg = 11.978
- pitch.peak_lateral_force_N = 10.0987
- pitch.peak_blade_moment_Nm = 2.2058
- pitch.peak_link_force_N = 105.93
- packaging.mount_points = 4
- structure.centrifugal_load_N = 221.464
- structure.centrifugal_load_overspeed_N = 318.908
- structure.overspeed_factor = 1.2
- structure.blade_load_factor = 4.0
- structure.blade_root_bending_Nm = 0.8712
- structure.blade_allowable_Nm = 25.2528
- structure.blade_margin = 28.9863
- structure.blade_combined_margin = 2.8341
- structure.blade_combined_margin_overspeed = 1.9681
- structure.blade_attachment_allowable_N = 540.0
- structure.blade_attachment_margin = 2.4383
- structure.blade_attachment_margin_overspeed = 1.6933
- structure.shaft_torque_Nm = 1.41944
- structure.shaft_allowable_Nm = 24.9563
- structure.shaft_margin = 17.5818
- structure.shaft_bending_Nm = 2.22965
- structure.shaft_combined_margin = 9.442
- structure.motor_shaft_torque_Nm = 0.43608
- structure.pitch_link_load_N = 105.93
- structure.pitch_link_allowable_N = 349.727
- structure.pitch_link_margin = 3.3015
- structure.blade_windup_deg = 2.3509
- results.total_mass_g = 607.97
- results.mass_envelope_g = 580.05
- results.mass_g_conservative = 684.7
- results.weight_N = 5.9642
- results.thrust_to_weight = 3.018
- results.bom_bought_inr = 36970
- results.bom_tooling_inr = 28800
- results.bom_total_inr = 65770
- results.bom_longest_lead_weeks = 4